

Chapter1

#Density $\rho = \frac{m}{V}$

Chapter2

#Displacement $\Delta x = x_2 - x_1$

#Average velocity

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

#Average speed

$$S_{avg} = \frac{\text{total distance}}{\Delta t}$$

#Instantaneous Velocity

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

#Average acceleration $a_{avg} = \frac{\Delta v}{\Delta t}$

#Instantaneous acceleration

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

#Constant acceleration

$$v = v_0 + at$$

$$x - x_0 = v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_0 + v)t$$

$$x - x_0 = v_0 t - \frac{1}{2} at^2$$

Chapter3

#Components of a Vector

$$a_x = a \cos \theta, \quad a_y = a \sin \theta$$

$$a = \sqrt{a_x^2 + a_y^2}, \quad \tan \theta = \frac{a_y}{a_x}$$

#The scalar product $\vec{a} \cdot \vec{b} = ab \cos \phi$

#The vector product $\vec{c} = \vec{a} \times \vec{b}$
 $c = ab \sin \phi$

Chapter4

Position vector $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

#Displacement $\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$

#Average velocity $\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t}$

#Instantaneous velocity $\vec{v} = \frac{d\vec{r}}{dt}$

#Average acceleration $\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t}$

#Instantaneous acceleration $\vec{a} = \frac{d\vec{v}}{dt}$

#Trajectory of projectile motion

$$y = (\tan \theta_0)x - \frac{gx^2}{2(v_0 \cos \theta_0)^2}$$

#Horizontal range $R = \frac{v_0^2}{g} \sin 2\theta_0$

#Uniform circular motion

$$a = \frac{v^2}{r} \quad T = \frac{2\pi r}{v}$$

#Relative motion $\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$

Chapter5

Newton's second law $\vec{F}_{net} = m\vec{a}$

#A gravitational force $F_g = mg$

#The weight of a body $W = mg$

#Newton's third law $\vec{F}_{BC} = -\vec{F}_{CB}$

Chapter6

#The maximum static friction force

$$f_{s,max} = \mu_s F_N$$

#The kinetic friction force

$$f_k = \mu_k F_N$$

#Drag force $D = \frac{1}{2} C \rho A v^2$

#Terminal speed $v_t = \sqrt{\frac{2F_g}{C \rho A}}$

#Uninform circular motion

$$a = \frac{v^2}{R} \quad F = m \frac{v^2}{R}$$

Chapter7

#Kinetic energy $K = \frac{1}{2} mv^2$

#Work done by a constant force

$$W = Fd \cos \phi = \vec{F} \cdot \vec{d}$$

#Work-kinetic energy theorem

$$\Delta K = K_f - K_i = W$$

#Work done by the gravitational force

$$W_g = mgd \cos \theta$$

#Work done in lifting and lowering an

object $\Delta K = K_f - K_i = W_a + W_g$

#Spring force

$$\vec{F}_s = -k\vec{d} \quad F_x = -kx$$

#Work done by a spring force

$$W_s = \frac{1}{2} kx_i^2 - \frac{1}{2} kx_f^2$$

#Work done by a variable force

$$W = \int_{x_i}^{x_f} F_x dx + \int_{y_i}^{y_f} F_y dy + \int_{z_i}^{z_f} F_z dz$$

#Power $P_{avg} = \frac{W}{\Delta t} \quad P = \frac{dW}{dt}$

$$P = Fv \cos \phi = \vec{F} \cdot \vec{v}$$

Chapter8

Potential energy

$$\Delta U = -W \quad \Delta U = -\int_{x_i}^{x_f} F(x) dx$$

#Gravitational potential energy

$$\Delta U = mg(y_f - y_i) = mg\Delta y$$

$$U(y) = mgy$$

#Elastic potential energy

$$U(x) = \frac{1}{2} kx^2$$

#Mechanical energy $E_{mec} = K + U$

#Principle of conservation of mechanical energy

$$K_2 + U_2 = K_1 + U_1$$

$$\Delta E_{mec} = \Delta K + \Delta U = 0$$

#Potential energy curves

$$F(x) = -\frac{dU(x)}{dx}$$

$$K(x) = E_{mec} - U(x)$$

#Work done on a system by an external force

$$W = \Delta E_{mec} = \Delta K + \Delta U$$

$$W = \Delta E_{mec} + \Delta E_{th}$$

$$\Delta E_{th} = f_k d$$

#Conservation of energy

$$W = \Delta E = \Delta E_{mec} + \Delta E_{th} + \Delta E_{int}$$

$$\text{\#Power } P_{avg} = \frac{\Delta E}{\Delta t} \quad P = \frac{dE}{dt}$$

Chapter9

$$\text{\#Center of mass } \vec{r}_{com} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$$

$$x_{com} = \frac{1}{M} \sum_{i=1}^n m_i x_i,$$

$$y_{com} = \frac{1}{M} \sum_{i=1}^n m_i y_i,$$

$$z_{com} = \frac{1}{M} \sum_{i=1}^n m_i z_i$$

#Newton's second law for a system

$$\vec{F}_{net} = M\vec{a}_{com}$$

#Linear momentum and newton's second law

$$\vec{p} = m\vec{v}, \quad \vec{F}_{net} = \frac{d\vec{p}}{dt}$$

$$\vec{P} = M\vec{v}_{com}, \quad \vec{F}_{net} = \frac{d\vec{P}}{dt}$$

#Collision and impulse

$$\vec{p}_f - \vec{p}_i = \Delta\vec{p} = \vec{J}$$

$$\vec{J} = \int_{t_i}^{t_f} \vec{F}(t) dt, \quad J = F_{avg} \Delta t$$

$$F_{avg} = -\frac{n}{\Delta t} \Delta p = -\frac{n}{\Delta t} m \Delta v$$

$$F_{avg} = -\frac{\Delta m}{\Delta t} \Delta v$$

#Conservation of linear momentum

$$\vec{P} = \text{constant}, \quad \vec{P}_i = \vec{P}_f$$

#Inelastic collision in one dimension

$$\vec{p}_{1i} + \vec{p}_{2i} = \vec{p}_{1f} + \vec{p}_{2f}$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

#Elastic collisions in one dimension

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i},$$

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$$

#Collisions in two dimensions

$$\vec{P}_{1i} + \vec{P}_{2i} = \vec{P}_{1f} + \vec{P}_{2f}$$

$$K_{1i} + K_{2i} = K_{1f} + K_{2f}$$

#Variable-mass system

$$Rv_{rel} = Ma, \quad v_f - v_i = v_{rel} \ln \frac{M_i}{M_f}$$

Chapter10

#Angular velocity and speed

$$\omega_{avg} = \frac{\Delta\theta}{\Delta t}, \quad \omega = \frac{d\theta}{dt}$$

#Angular acceleration

$$\alpha_{avg} = \frac{\Delta\omega}{\Delta t}, \quad \alpha = \frac{d\omega}{dt}$$

#The kinetic equations for constant angular acceleration

$$\omega = \omega_0 + \alpha t$$

$$\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

$$\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$$

$$\theta - \theta_0 = \omega_0 t - \frac{1}{2} \alpha t^2$$

#Linear and angular variables related

$$s = \theta r, \quad v = \omega r, \quad a_t = \alpha r,$$

$$\alpha_r = \frac{v^2}{r} = \omega^2 r, \quad T = \frac{2\pi r}{v} = \frac{2\pi}{\omega}$$

#Rotational kinetic energy and rotational inertia

$$K = \frac{1}{2} I \omega^2$$

$$I = \sum m_i r_i^2, \quad I = \int r^2 dm$$

#The parallel-axis theorem

$$I = I_{com} + Mh^2$$

#Torque $\tau = rF_t = r_\perp F = rF \sin \phi$

#Newton's second law in angular form

$$\tau_{net} = I\alpha$$

#Work and rotational kinetic energy

$$W = \int_{\theta_i}^{\theta_f} \tau d\theta, \quad P = \frac{dW}{dt} = \tau\omega$$

$$W = \tau(\theta_f - \theta_i)$$

$$\Delta K = K_f - K_i = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2 = W$$

Chapter11

#Rolling bodies

$$v_{com} = \omega R, \quad a_{com} = \alpha R,$$

$$K = \frac{1}{2} I_{com} \omega^2 + \frac{1}{2} M v_{com}^2$$

$$a_{com,x} = -\frac{g \sin \theta}{1 + I_{com}/MR^2}$$

#Torque as a vector

$$\vec{\tau} = \vec{r} \times \vec{F}$$

#Angular momentum of a particle

$$\vec{l} = \vec{r} \times \vec{p} = \vec{r} \times (m\vec{v})$$

$$l = rmv \sin \phi$$

#Newton's second law in angular form

$$\vec{\tau}_{net} = \frac{d\vec{l}}{dt}$$

#Angular momentum of a system

$$\vec{L} = \vec{l}_1 + \vec{l}_2 + \cdots + \vec{l}_n = \sum_{i=1}^n \vec{l}_i$$

$$\vec{\tau}_{net} = \frac{d\vec{L}}{dt}$$

#Angular momentum of rigid body

$$L = I\omega$$

#Conservation of angular momentum

$$\vec{L} = \text{constant}, \quad \vec{L}_i = \vec{L}_f$$

#Precession of a gyroscope

$$\Omega = \frac{Mgr}{I\omega}$$

Chapter12

#Static equilibrium

$$\vec{F}_{net} = 0, \quad F_{net,x} = 0, \quad F_{net,y} = 0$$

$$\vec{\tau}_{net} = 0, \quad \tau_{net,z} = 0$$

#Elastic moduli

$$\frac{F}{A} = E \frac{\Delta L}{L}, \quad \frac{F}{A} = G \frac{\Delta x}{L},$$

$$p = B \frac{\Delta V}{V}$$

Chapter13

#The law of gravitation

$$F = G \frac{m_1 m_2}{r^2}$$

Superposition

$$\vec{F}_{1,net} = \sum_{i=1}^n \vec{F}_{1i}, \quad \vec{F}_1 = \int d\vec{F}$$

#Gravitational acceleration

$$F = ma_g, \quad a_g = \frac{GM}{r^2}$$

#Gravitation with a spherical shell

$$F = \frac{GmM}{R^3} r$$

#Gravitational potential energy

$$U = -\frac{GMm}{r}$$

#Potential energy of a system

$$U = -\left(\frac{Gm_1 m_2}{r_{12}} + \frac{Gm_1 m_3}{r_{13}} + \frac{Gm_2 m_3}{r_{23}}\right)$$

#Escape speed

$$v = \sqrt{\frac{2GM}{R}}$$

#Law of periods

$$T^2 = \left(\frac{4\pi^2}{GM}\right)r^3$$

#Energy in planetary motion

$$U = -\frac{GMm}{r}, \quad K = \frac{GMm}{2r}$$

$$E = K + U$$

$$E = -\frac{GMm}{2r}, \quad E = -\frac{GMm}{2a}$$

Chapter15

$$\# \text{ Period } T = \frac{1}{f}, \quad \omega = \frac{2\pi}{T} = 2\pi f$$

#Simple harmonic motion

$$x = x_m \cos(\omega t + \phi)$$

$$v = -\omega x_m \sin(\omega t + \phi)$$

$$a = -\omega^2 x_m \cos(\omega t + \phi)$$

#The linear oscillator

$$\omega = \sqrt{\frac{k}{m}}, \quad T = 2\pi \sqrt{\frac{m}{k}}$$

#Pendulums

$$T = 2\pi \sqrt{I/\kappa}, \quad T = 2\pi \sqrt{L/g}$$

$$T = 2\pi \sqrt{I/mgh}$$

#Damped harmonic motion

$$x(t) = x_m e^{-bt/2m} \cos(\omega' t + \phi)$$

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

$$E(t) \approx \frac{1}{2} k x_m^2 e^{-bt/m}$$

#Forced oscillations and resonance

$$\omega_d = \omega$$

Chapter16

#Sinusoidal waves

$$y(x, t) = y_m \sin(kx - \omega t)$$

$$k = \frac{2\pi}{\lambda}, \quad \frac{\omega}{2\pi} = f = \frac{1}{T}$$

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f$$

#Equation of a traveling wave

$$y(x, t) = h(kx \pm \omega t)$$

#Wave speed on stretched string

$$v = \sqrt{\frac{\tau}{\mu}}$$

$$\# \text{ Power } P_{avg} = \frac{1}{2} \mu v \omega^2 y_m^2$$

#Interference of waves

$$y'(x, t) = [2y_m \cos(\frac{1}{2}\phi)] \sin(kx - \omega t + \frac{1}{2}\phi)$$

#Standing waves

$$y'(x, t) = [2y_m \sin kx] \cos \omega t$$

#Resonance

$$f = \frac{v}{\lambda} = n \frac{v}{2L}, \quad \text{for } n=1, 2, 3, \dots$$

Chapter17

#Sound waves

$$s = s_m \cos(kx - \omega t)$$

$$\Delta p = \Delta p_m \sin(kx - \omega t)$$

$$\Delta p_m = (v\rho\omega)s_m, \quad v = \sqrt{\frac{B}{\rho}}$$

$$\# \text{ Interference } \phi = \frac{\Delta L}{\lambda} 2\pi$$

#Fully constructive interference

$$\phi = m(2\pi), \quad \frac{\Delta L}{\lambda} = 0, 1, 2, \dots$$

#Fully destructive interference

$$\phi = (2m+1)\pi,$$

$$\frac{\Delta L}{\lambda} = 0.5, 1.5, 2.5, \dots$$

#Sound intensity

$$I = \frac{P}{A}, \quad I = \frac{1}{2} \rho v \omega^2 s_m^2, \quad I = \frac{P_s}{4\pi r^2}$$

#Sound level in decibels

$$\beta = (10\text{dB}) \log \frac{I}{I_0},$$

$$I_0 = 10^{-12} \text{ W/m}^2$$

#Standing wave patterns in pipe

$$f = \frac{v}{\lambda} = \frac{nv}{2L}, \quad n=1,2,3,\dots$$

$$f = \frac{v}{\lambda} = \frac{nv}{4L}, \quad n=1,3,5,\dots$$

#Beats $f_{beat} = f_1 - f_2$

#The Doppler effect

$$f' = f \frac{v \pm v_D}{v \pm v_S}$$

#Shock wave $\sin \theta = \frac{v}{v_S}$

Chapter 18

#The Kelvin temperature scale

$$T = (273.16K) \left(\lim_{gas \rightarrow 0} \frac{p}{p_3} \right)$$

#Celsius and Fahrenheit scales

$$T_C = T - 273.15^0, \quad T_F = \frac{9}{5} T_C + 32^0$$

#Thermal expansion $\beta = 3\alpha$

$$\Delta L = L\alpha\Delta T, \quad \Delta V = V\beta\Delta T,$$

#Heat capacity and specific heat

$$Q = C(T_f - T_i), \quad Q = cm(T_f - T_i)$$

#Heat of transformation $Q = Lm$

#Work associated with volume change

$$W = \int dW = \int_{V_i}^{V_f} p dV$$

#First law of thermodynamics

$$\Delta E_{int} = E_{int,f} - E_{int,i} = Q - W$$

$$dE_{int} = dQ - dW$$

#Conduction $P_{con} = \frac{Q}{t} = kA \frac{T_H - T_C}{L}$

#Radiation

$$P_{rad} = \sigma \varepsilon A T^4, \quad P_{abs} = \sigma \varepsilon A T_{env}^4$$

Chapter 19

#Avogadro's number $M = mN_A$

$$n = \frac{N}{N_A} = \frac{M_{sam}}{M} = \frac{M_{sam}}{mN_A}$$

#Ideal gas

$$PV = nRT, \quad PV = NkT$$

#Work in an isothermal volume change

$$W = nRT \ln \frac{V_f}{V_i}$$

#Pressure, temperature, and molecular

speed $p = \frac{nMv_{rms}^2}{3V}, \quad v_{rms} = \sqrt{\frac{3RT}{M}}$

#Temperature and kinetic energy

$$K_{avg} = \frac{3}{2} kT$$

#Mean free path $\lambda = \frac{1}{\sqrt{2}\pi d^2 N/V}$

#Maxwell speed distribution

$$P(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/2RT}$$

$$v_{avg} = \sqrt{\frac{8RT}{\pi M}}, \quad v_p = \sqrt{\frac{2RT}{M}}$$

#Molar specific heats

$$Q = nC_V \Delta T, \quad Q = nC_p \Delta T$$

$$\Delta E_{int} = nC_V \Delta T, \quad E_{int} = nC_V T$$

$$C_p = C_V + R$$

#Degrees of freedom and C_V

$$C_V = \frac{f}{2} R$$

#Adiabatic process

$$PV^\gamma = \text{a constant}, \quad \gamma = C_p / C_V$$

Chapter 20

#Calculating entropy change

$$\Delta S = S_f - S_i = \int_i^f \frac{dQ}{T}$$

$$\Delta S = S_f - S_i = \frac{Q}{T}$$

$$\Delta S = S_f - S_i \approx \frac{Q}{T}$$

$$\Delta S = S_f - S_i = nR \ln \frac{V_f}{V_i} + nC_V \ln \frac{T_f}{T_i}$$

#The second law of thermodynamic

$$\Delta S \geq 0$$

#Engines

$$\varepsilon = \frac{\text{energy we get}}{\text{energy we pay for}} = \frac{|W|}{|Q_H|}$$

#A Carnot engine

$$\varepsilon = 1 - \frac{|Q_L|}{|Q_H|} = 1 - \frac{T_L}{T_H}$$

#Refrigerators

$$K = \frac{\text{what we want}}{\text{what we pay for}} = \frac{|Q_L|}{|W|}$$

#A Carnot refrigerator

$$K_C = \frac{|Q_L|}{|Q_H| - |Q_L|} = \frac{T_L}{T_H - T_L}$$

#Entropy from a statistics view

$$W = \frac{N!}{n_1! n_2!}, \quad S = k \ln W$$

Constants

#Speed of light $c = 3.00 \times 10^8 \text{ m/s}$

#Free-fall acceleration $g = 9.81 \text{ m/s}^2$

#Gravitational constant

$$G = 6.67 \times 10^{-11} \text{ m}^3/\text{s}^2 \cdot \text{kg}$$

#Universal gas constant

$$R = 8.31 \text{ J/mol} \cdot \text{K}$$

#Avogadro constant

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

#Boltzmann constant

$$k = 1.38 \times 10^{-23} \text{ J/K}$$

#Electron mass

$$m_e = 9.109 \times 10^{-31} \text{ kg}$$

#Proton mass $m_p = 1.673 \times 10^{-27} \text{ kg}$

#Neutron mass

$$m_n = 1.675 \times 10^{-27} \text{ kg}$$

#Mass of the sun $1.99 \times 10^{30} \text{ kg}$

#Mass of the earth $5.98 \times 10^{24} \text{ kg}$

#Mass of the moon $7.36 \times 10^{22} \text{ kg}$

#Mean radius of the sun $6.96 \times 10^8 \text{ m}$

#Mean radius of the earth

$$6.37 \times 10^6 \text{ m}$$

#Mean radius of the moon

$$1.74 \times 10^6 \text{ m}$$

#Some rotational inertias

(a) Hoop about central axis $I = MR^2$

(b) Annular cylinder (or ring) about

central axis $I = \frac{1}{2} M (R_1^2 + R_2^2)$

(c) Solid cylinder (or disk) about

central axis $I = \frac{1}{2} MR^2$

(d) Solid cylinder (or disk) about

central diameter $I = \frac{1}{4} MR^2 + \frac{1}{12} ML^2$

(e) Thin rod about axis through center

perpendicular to length $I = \frac{1}{12} ML^2$

(f) Solid sphere about any diameter

$$I = \frac{2}{5} MR^2$$

(g) Thin spherical shell about any

diameter $I = \frac{2}{3} MR^2$

(h) Hoop about any diameter

$$I = \frac{1}{2} MR^2$$

(i) Slab about perpendicular axis

through center $I = \frac{1}{12} M (a^2 + b^2)$

Chapter 21

#Electric current $i = \frac{dq}{dt}$

#Coulomb's law $F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$

Chapter 22

#Definition of Electric field $\vec{E} = \frac{\vec{F}}{q_0}$

#Field Due to a Point Charge

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

#Field Due to an Electric Dipole

$$E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}$$

#Field Due to Charged disk

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right)$$

#Force on a Point charge in an Electric

Field $\vec{F} = q\vec{E}$

#Dipole in an Electric Field

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad U = -\vec{p} \cdot \vec{E}$$

Chapter 23

#Gauss' law

$$\epsilon_0 \Phi = q_{enc}, \quad \Phi = \oint \vec{E} \cdot d\vec{A}$$

#conducting surface $E = \frac{\sigma}{\epsilon_0}$

#line of charge $E = \frac{\lambda}{2\pi\epsilon_0 r}$

#sheet of charge $E = \frac{\sigma}{2\epsilon_0}$

#spherical shell, for $r > R$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

#spherical shell, for $r < R$ $E = 0$

#A uniform sphere of charge

$$E = \left(\frac{q}{4\pi\epsilon_0 R^3}\right)r$$

Chapter 24

#Electric potential $V = \frac{-W_\infty}{q_0} = \frac{U}{q_0}$

#Electric potential energy

$$U = qV, \Delta U = q\Delta V = q(V_f - V_i)$$

#Mechanical energy

$$\Delta K = -q\Delta V, \Delta K = -q\Delta V + W_{app}$$

#Finding V from E

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$$

$$V = -\int_i^f \vec{E} \cdot d\vec{s}, \quad \Delta V = -E\Delta x$$

#Potential Due to a Charged Particle

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}, \quad V = \sum_{i=1}^n V_i = \sum_{i=1}^n \frac{q_i}{r_i}$$

#Potential Due to an Electric Dipole

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad (r \gg d)$$

#Potential Due to a Continuous Charge

Distribution $V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$

#Calculating E from V

$$E_s = -\frac{\partial V}{\partial s}, \quad E = -\frac{\Delta V}{\Delta s}$$

#Potential energy for two particles

$$U = W = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

Chapter 25

#Definition of Capacitance

$$q = CV$$

#A parallel-plate capacitor $C = \frac{\epsilon_0 A}{d}$

#A cylindrical capacitor

$$C = 2\pi\epsilon_0 \frac{L}{\ln(b/a)}$$

#A spherical capacitor

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}$$

#An isolated sphere $C = 4\pi\epsilon_0 R$

#capacitors in parallel $C_{eq} = \sum_{j=1}^n C_j$

#capacitors in series $\frac{1}{C_{eq}} = \sum_{j=1}^n \frac{1}{C_j}$

#Potential energy $U = \frac{q^2}{2C} = \frac{1}{2} CV^2$

#Energy density $u = \frac{1}{2} \epsilon_0 E^2$

#Gauss' Law with a Dielectric

$$\epsilon_0 \oint \kappa \vec{E} \cdot d\vec{A} = q$$

Chapter 26

#Electric current $i = \frac{dq}{dt}$

#Current density $i = \int \vec{J} \cdot d\vec{A}$

#Drift speed $\vec{J} = (ne)\vec{v}_d$

#Definition of R $R = \frac{V}{i}$

#Definitions of ρ and σ

$$\rho = \frac{1}{\sigma} = \frac{E}{J}$$

$\vec{E} = \rho \vec{J}$, $R = \rho \frac{L}{A}$

#Change of ρ with temperature

$$\rho - \rho_0 = \rho_0 \alpha (T - T_0)$$

#Resistivity of a metal $\rho = \frac{m}{e^2 n \tau}$

#Power $P = iV$

#Resistivity dissipation $P = i^2 R = \frac{V^2}{R}$

Chapter 27

Emf $\mathcal{E} = \frac{dW}{dq}$

#Single loop circuit $i = \frac{\mathcal{E}}{R+r}$

#Power

$$P = iV, \quad P_r = i^2 r, \quad P_{emf} = i\mathcal{E}$$

#Resistances in series $R_{eq} = \sum_{j=1}^n R_j$

#Resistance in parallel $\frac{1}{R_{eq}} = \sum_{j=1}^n \frac{1}{R_j}$

#Charging a capacitor, $RC = \tau$

$$q = C\mathcal{E}(1 - e^{-t/RC})$$

$$i = \frac{dq}{dt} = \left(\frac{\mathcal{E}}{R}\right)e^{-t/RC}$$

#Discharging a capacitor, $RC = \tau$

$$q = q_0 e^{-t/RC}$$

$$i = \frac{dq}{dt} = -\left(\frac{q_0}{RC}\right)e^{-t/RC}$$

Chapter 28

#Magnetic field $\vec{B}$ $\vec{F}_B = q\vec{v} \times \vec{B}$

#A charged particle circulating in a magnetic field

$$|q|vB = \frac{mv^2}{r}, \quad r = \frac{mv}{qB}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{T} = \frac{|q|B}{2\pi m}$$

#The Hall Effect

$$n = \frac{Bi}{Vle}, \quad V = vBd$$

#Magnetic force

$$\vec{F}_B = i\vec{L} \times \vec{B}, \quad d\vec{F}_B = id\vec{L} \times \vec{B}$$

#Magnetic dipole moment

$$\mu = NiA, \quad \vec{\tau} = \vec{\mu} \times \vec{B}$$

#Orientation energy of a Magnetic

$$\text{dipole } U(\theta) = -\vec{\mu} \cdot \vec{B}$$

#Work done on dipole by the agent

$$W_a = \Delta U = U_f - U_i$$

Chapter 29

#The Biot-Savart Law

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{s} \times \hat{r}}{r^2}$$

#A long straight wire $B = \frac{\mu_0 i}{2\pi r}$

#A circular Arc $B = \frac{\mu_0 i \phi}{4\pi R}$

#Force between parallel currents

$$F_{ba} = i_b L B_a \sin 90^\circ = \frac{\mu_0 L i_a i_b}{2\pi d}$$

#Ampere's Law

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$$

#Field of an ideal solenoid $B = \mu_0 i n$

#Field of an ideal toroid $B = \frac{\mu_0 i N}{2\pi r}$

#Field of a magnetic dipole

$$\vec{B}(z) = \frac{\mu_0}{2\pi} \frac{\vec{\mu}}{z^3}$$

Chapter 30

#Magnetic flux $\Phi_B = \int \vec{B} \cdot d\vec{A}$

#Faraday's Law $\mathcal{E} = -\frac{d\Phi_B}{dt}$

#Faraday's Law (N turns)

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

#Emf $\mathcal{E} = \oint \vec{E} \cdot d\vec{s}$

#The induced electric field

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

#Definition of inductance $L = \frac{N\Phi_B}{i}$

#Inductance of solenoid $\frac{L}{l} = \mu_0 n^2 A$

#Self-induction $\mathcal{E}_L = -L \frac{di}{dt}$

#Series RL Circuit $\tau_L = \frac{L}{R}$

$$i = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau_L}) \quad (\text{rise of current})$$

$$i = i_0 e^{-t/\tau_L} \quad (\text{decay of current})$$

#Magnetic energy $U_B = \frac{1}{2} Li^2$

#Magnetic energy density $u_B = \frac{B^2}{2\mu_0}$

#Mutual induction

$$\mathcal{E}_2 = -M \frac{di_1}{dt}, \quad \mathcal{E}_1 = -M \frac{di_2}{dt}$$

Chapter 31

#LC Oscillations

$$L \frac{d^2 q}{dt^2} + \frac{1}{C} q = 0, \quad \omega = \frac{1}{\sqrt{LC}}$$

$$q = Q \cos(\omega t + \phi),$$

$$i = -\omega Q \sin(\omega t + \phi)$$

#Damped Oscillations

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0$$

$$q = Q e^{-Rt/2L} \cos(\omega' t + \phi),$$

$$\omega' = \sqrt{\omega^2 - (R/2L)^2}$$

#Forced Oscillations

$$\mathcal{E} = \mathcal{E}_m \sin(\omega_d t); i = I \sin(\omega_d t - \phi)$$

#Resonance

Maximum $I = \frac{\varepsilon_m}{R}$, $\phi = 0$,

$$\omega_d = \omega = \frac{1}{\sqrt{LC}}, \quad X_C = X_L,$$

#Single circuit elements

$$V_R = IR, \quad V_C = IX_C, \quad X_C = \frac{1}{\omega_d C},$$

$$V_L = IX_L, \quad X_L = \omega_d L$$

#Series RLC circuits

$$I = \frac{\varepsilon_m}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{\varepsilon_m}{\sqrt{R^2 + (\omega_d L - 1/\omega_d C)^2}}$$

$$\tan \phi = \frac{X_L - X_C}{R} \quad (\text{phase constant})$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad (\text{impedance})$$

#Power $I_{rms} = I/\sqrt{2}$

$$V_{rms} = V/\sqrt{2} \quad \varepsilon_{rms} = \varepsilon_m/\sqrt{2}$$

$$P_{avg} = I_{rms}^2 R = \varepsilon_{rms} I_{rms} \cos \phi$$

#Transformers

$$V_s = V_p \frac{N_s}{N_p}, \quad I_s = I_p \frac{N_p}{N_s},$$

$$R_{eq} = \left(\frac{N_p}{N_s}\right)^2$$

Chapter32

#Gauss' Law for magnetic field

$$\Phi_B = \oint \vec{B} \cdot d\vec{A} = 0$$

#Maxwell's law of induction

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

#Ampere-Maxwell law

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{enc}$$

#Displacement current

$$i_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Chapter33

#Electromagnetic waves

$$E = E_m \sin(kx - \omega t)$$

$$B = B_m \sin(kx - \omega t)$$

$$c = \frac{E}{B} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$$\omega = 2\pi f, \quad v = \lambda f$$

#Energy flow

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}, \quad E_{rms} = E_m/\sqrt{2}$$

$$I = \frac{1}{c\mu_0} E_{rms}^2, \quad I = \frac{P_s}{4\pi r^2}$$

#Total absorption

$$F = \frac{IA}{c}, \quad p_r = \frac{I}{c}$$

#Total reflection

$$F = \frac{2IA}{c}, \quad p_r = \frac{2I}{c}$$

#Polarizing sheets

$$I = \frac{1}{2} I_0, \quad I = I_0 \cos^2 \theta$$

#Law of refraction

$$n_2 \sin \theta_2 = n_1 \sin \theta_1$$

#Critical angle $\theta_c = \sin^{-1}(\frac{n_2}{n_1})$

#Brewster angle $\theta_B = \tan^{-1}(\frac{n_2}{n_1})$

Chapter35

#Wavelength and index of refraction

$$\lambda_n = \frac{\lambda}{n}$$

#Young's experiment (for m=0,1,2,...)

$$\text{Maxima } d \sin \theta = m\lambda$$

$$\text{Minima } d \sin \theta = (m + \frac{1}{2})\lambda$$

#Intensity in Two-slit interference

$$I = 4I_0 \cos^2 \frac{1}{2} \phi, \quad \phi = \frac{2\pi d}{\lambda} \sin \theta$$

#Thin-film interference

$$\text{Maxima } 2L = (m + \frac{1}{2}) \frac{\lambda}{n_2}$$

$$\text{Minima } 2L = m \frac{\lambda}{n_2}$$

Chapter36

#Single-slit diffraction m=1,2,3,...

$$\text{Minima } a \sin \theta = m\lambda$$

$$I(\theta) = I_m \left(\frac{\sin \alpha}{\alpha}\right)^2, \quad \alpha = \frac{\pi a}{\lambda} \sin \theta$$

#Circular aperture diffraction

$$\text{First minimum } \sin \theta = \frac{1.22\lambda}{d}$$

#Rayleigh's criterion $\theta_R = 1.22 \frac{\lambda}{d}$

#Double-slit diffraction

$$I(\theta) = I_m (\cos^2 \beta) \left(\frac{\sin \alpha}{\alpha}\right)^2$$

$$\beta = (\pi d/\lambda) \sin \theta, \quad \alpha \text{ as single-slit}$$

#Diffraction gratings

$$d \sin \theta = m\lambda \quad \text{for } m=0,1,2,3,\dots$$

#Half width $\Delta\theta_{hw} = \frac{\lambda}{Nd \cos \theta}$

#Dispersion $D = \frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d \cos \theta}$

#Resolving power $R = \frac{\lambda_{avg}}{\Delta\lambda} = Nm$

#Bragg's Law

$$2d \sin \theta = m\lambda, \quad m=1,2,3,\dots$$

Chapter37

#Time dilation $\Delta t = \gamma \cdot \Delta t_0$

$$\beta = v/c, \quad \gamma = 1/\sqrt{1-\beta^2}$$

#Length contraction $L = \frac{L_0}{\gamma}$

#The Lorentz transformation

$$x' = \gamma(x - vt), \quad y' = y, \quad z' = z,$$

$$t' = \gamma(t - vx/c^2)$$

#Relativity of velocities

$$u = \frac{u' + v}{1 + u'v/c^2}$$

#Relativistic Doppler Effect

$$\lambda = \lambda_0 \sqrt{\frac{1 + \beta}{1 - \beta}}, \quad v = \frac{|\Delta\lambda|}{\lambda_0}$$

(source and detector separating)

#Transvers Doppler Effect

$$f = f_0 \sqrt{1 - \beta^2}$$

#Momentum and energy

$$\vec{p} = \gamma m \vec{v}, \quad K = mc^2(\gamma - 1),$$

$$E = mc^2 + K = \gamma mc^2$$

$$(pc)^2 = K^2 + 2Kmc^2$$

$$E^2 = (pc)^2 + (mc^2)^2$$

$$Q = M_i c^2 - M_f c^2 = -\Delta M \cdot c^2$$

Chapter38

#Light quanta-photons

$$E = hf \quad p = \frac{hf}{c} = \frac{h}{\lambda}$$

#Photoelectric Effect

$$hf = K_{\max} + \Phi$$

#Compton shift

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$$

#Ideal blackbody radiation

$$S(\lambda) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

$$\lambda_{\max} T = 2898 \mu m \cdot K$$

#Matter waves $\lambda = \frac{h}{p}$

#The wave function

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}[E - U(x)]\psi = 0$$

#Heisenberg's uncertainty principle

$$\hbar = h/(2\pi)$$

$$\Delta x \cdot \Delta p_x \geq \hbar, \quad \Delta y \cdot \Delta p_y \geq \hbar, \quad \Delta z \cdot \Delta p_z \geq \hbar$$

#Potential step $T = 1 - R$

#Barrier tunneling $T \approx e^{-2bL}$

$$b = \sqrt{\frac{8\pi^2m(U_b - E)}{h^2}}$$

Chapter39

#Electron in an infinite potential well

$$E_n = \left(\frac{h^2}{8mL^2}\right)n^2, \quad \text{for } n = 1, 2, 3, \dots$$

$$\Delta E = E_{\text{high}} - E_{\text{low}}$$

$$hf = \frac{hc}{\lambda} = \Delta E = E_{\text{high}} - E_{\text{low}}$$

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right),$$

$$\text{for } n = 1, 2, 3, \dots$$

$$\int_{-\infty}^{+\infty} \psi_n^2(x) dx = 1$$

#Two dimensional electron trap

$$E_{n_x, n_y} = \frac{h^2}{8m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} \right)$$

$$\psi_{n_x, n_y} = \sqrt{\frac{2}{L_x}} \sin\left(\frac{n_x\pi}{L_x}x\right) \sqrt{\frac{2}{L_y}} \sin\left(\frac{n_y\pi}{L_y}y\right)$$

Constants

#Speed of light $c = 3.00 \times 10^8 \text{ m/s}$

#Elementary charge

$$e = 1.60 \times 10^{-19} \text{ C}$$

#Free-fall acceleration $g = 9.81 \text{ m/s}^2$

#Gravitational constant

$$G = 6.67 \times 10^{-11} \text{ m}^3/\text{s}^2 \cdot \text{kg}$$

#Universal gas constant

$$R = 8.31 \text{ J/mol} \cdot \text{K}$$

#Avogadro constant

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

#Boltzmann constant

$$k = 1.38 \times 10^{-23} \text{ J/K}$$

#Permittivity constant

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

#Permeability constant

$$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A} = 1.26 \times 10^{-6} \text{ T} \cdot \text{m/A}$$

#Planck constant

$$h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$$

#Electron mass

$$m_e = 9.109 \times 10^{-31} \text{ kg}$$

#Proton mass $m_p = 1.673 \times 10^{-27} \text{ kg}$

#Neutron mass

$$m_n = 1.675 \times 10^{-27} \text{ kg}$$

#

$$\# \int \frac{1}{x} dx = \ln|x| + C$$